Project: Worked examples
Title: EN 1993-1-1 member checks (basic)
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1. Scope and method

Resistance checks of a steel member of class 1, 2 or 3 cross-section: tension (6.2.3), compression (6.2.4) and flexural buckling about y and z (6.3.1), bending about y and z (6.2.5), shear with the high-shear reduction (6.2.6, 6.2.8), the linear cross-section interaction (6.2.1(7)) and a simplified member interaction (6.3.3 with user-entered k factors). Lateral-torsional buckling and class 4 sections are outside this sheet.

WHAT IT CALCULATES	RELATION
Axial plastic resistance	$N_{pl,Rd} = A f_y/\gamma_{M0}$
Bending resistance, plastic modulus for class 1–2 and elastic for class 3	$M_{c,Rd} = W f_y/\gamma_{M0}$
Shear resistance	$V_{pl,Rd} = A_v f_y/(\sqrt{3}\,\gamma_{M0})$
Bending resistance reduced by high shear, when $V_{Ed} > 0.5\,V_{pl,Rd}$	$\rho = (2V_{Ed}/V_{pl,Rd} - 1)^2$, resistance $\times (1 - \rho)$
Elastic critical load and non-dimensional slenderness	$N_{cr} = \pi^2 EI/L_{cr}^2$; $\bar{\lambda} = \sqrt{(A f_y/N_{cr})}$
Buckling reduction factor and buckling resistance	$\Phi = 0.5[1 + \alpha(\bar{\lambda} - 0.2) + \bar{\lambda}^2]$; $\chi = 1/(\Phi + \sqrt{(\Phi^2 - \bar{\lambda}^2)}) \le 1$; $N_{b,Rd} = \chi A f_y/\gamma_{M1}$
Cross-section check, linear interaction	$N_{Ed}/N_{Rd} + M_{y,Ed}/M_{y,Rd} + M_{z,Ed}/M_{z,Rd} \le 1$
Member check, axial force and bending with buckling	$N_{Ed}/N_{b,Rd,min} + k_{yy} M_{y,Ed}/M_{y,Rd} + k_{zz} M_{z,Ed}/M_{z,Rd} \le 1$

Basis: Tension[1](6.2.3); Compression[1](6.2.4); Bending moment[1](6.2.5); Shear[1](6.2.6); Bending and shear[1](6.2.8); Linear summation of utilisations[1](6.2.1(7)); Uniform members in compression: buckling resistance, buckling curves[1](6.3.1.1, 6.3.1.2); Uniform members in bending and axial compression[1](6.3.3); Imperfection factors for the buckling curves[2](§6.3.1).

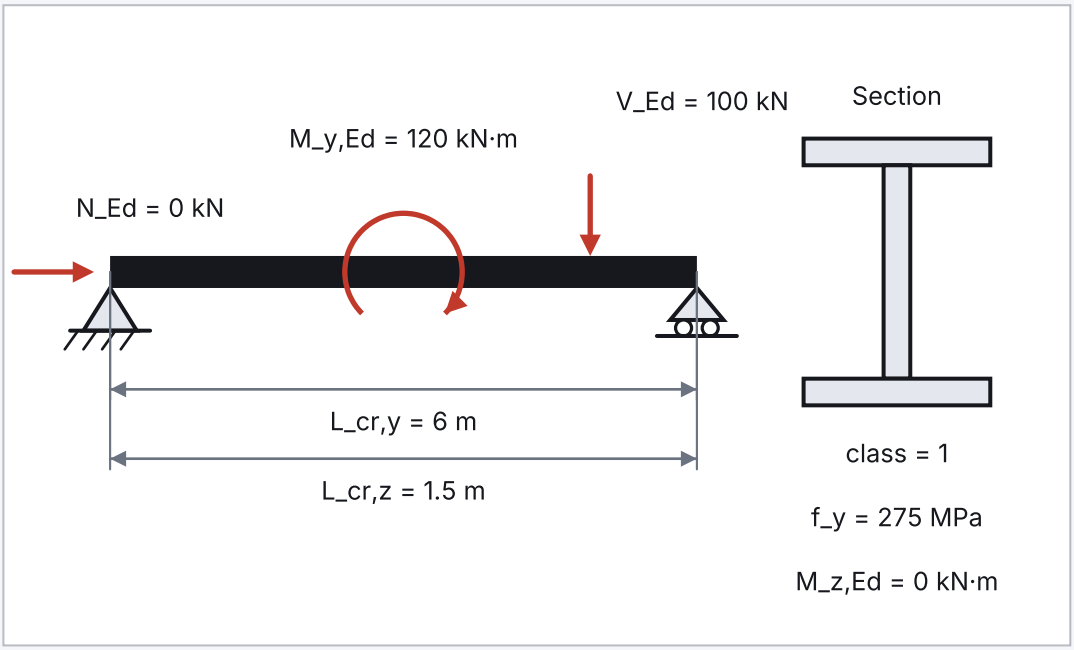


Figure 1 — A steel member under axial force, bending and shear, its buckling lengths about both axes, and its section. Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
N_Ed	Design axial force (tension +, compression −)	0	kN	reference: S2 frame sheet, two-span beam, ULS	-100000 … 100000 kN
M_y_Ed	Design moment about y-y (major)	120	kN·m	reference: S2 frame sheet, two-span beam, ULS (rounded)	0 … 100000 kN·m
M_z_Ed	Design moment about z-z (minor)	0	kN·m	reference: S2 frame sheet	0 … 100000 kN·m
V_Ed	Design shear force	100	kN	reference: S2 frame sheet, two-span beam, ULS (rounded)	0 … 100000 kN
f_y	Yield strength (for the thickness)	275	MPa	spec: BS EN 10025-2 S275, t ≤ 16 mm	200 … 700 MPa
E	Modulus of elasticity	210	GPa	spec: BS EN 1993-1-1 3.2.6	190 … 220 GPa
gamma_M0	Partial factor γ_M0 (cross-section resistance)	1		spec: UK NA to BS EN 1993-1-1 NA.2.15	1 … 1.25
gamma_M1	Partial factor γ_M1 (member buckling)	1		spec: UK NA to BS EN 1993-1-1 NA.2.15	1 … 1.25
section_class	Cross-section class	1		assumption: UB 305×165×40 S275 is class 1 in bending (Blue Book).	any

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
A	Area A	51.3	cm^2	reference: S1 section sheet / Blue Book UB 305×165×40	1 … 100000 cm^2
W_pl_y	W_pl,y	623	cm^3	reference: S1 section sheet / Blue Book	0.1 … 10000000 cm^3
W_el_y	W_el,y	560	cm^3	reference: S1 section sheet / Blue Book	0.1 … 10000000 cm^3
W_pl_z	W_pl,z	142	cm^3	reference: S1 section sheet / Blue Book	0.1 … 10000000 cm^3
W_el_z	W_el,z	92.6	cm^3	reference: S1 section sheet / Blue Book	0.1 … 10000000 cm^3
I_y	I_y	8503	cm^4	reference: S1 section sheet / Blue Book	0.1 … 1000000000 cm^4
I_z	I_z	764	cm^4	reference: S1 section sheet / Blue Book	0.1 … 1000000000 cm^4
h	Depth h (shear-area helper)	303.4	mm	spec: BS EN 10365 UB 305×165×40	1 … 5000 mm
b	Width b (helper)	165	mm	spec: BS EN 10365 UB 305×165×40	1 … 5000 mm
t_w	Web thickness (helper)	6	mm	spec: BS EN 10365 UB 305×165×40	0.1 … 500 mm
t_f	Flange thickness (helper)	10.2	mm	spec: BS EN 10365 UB 305×165×40	0.1 … 500 mm
r	Root radius (helper)	8.9	mm	spec: BS EN 10365 UB 305×165×40	0 … 500 mm
eta	Shear-area factor η	1		spec: UK NA to BS EN 1993-1-5 NA.2.4	1 … 1.2
L_cr_y	Buckling length about y-y	6	m	assumption: Span 6 m.	0.1 … 100 m
L_cr_z	Buckling length about z-z	1.5	m	assumption: Compression flange restrained at 1.5 m by the deck.	0.1 … 100 m
curve_y	Buckling curve about y-y	a		spec: BS EN 1993-1-1 Table 6.2: rolled, h/b > 1.2, t_f ≤ 40, y-y	any
curve_z	Buckling curve about z-z	b		spec: BS EN 1993-1-1 Table 6.2: rolled, h/b > 1.2, t_f ≤ 40, z-z	any

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
k_yy	Interaction factor k_yy (member check)	1		assumption: No axial force; interaction factors irrelevant.	0.5 … 2
k_zz	Interaction factor k_zz (member check)	1		assumption: No axial force; interaction factors irrelevant.	0.5 … 2

3. Assumptions

A1. The cross-section class is as stated by the user (1, 2 or 3); class 4 effective properties are not derived here. (6.2.2.5 effective widths are outside this sheet.)
A2. Lateral-torsional buckling is not checked: the compression flange is assumed restrained, or LTB is checked elsewhere. (6.3.2 not implemented.)
A3. The member interaction uses the user's k_yy and k_zz (default 1.0) with the smaller flexural-buckling reduction on the axial term; the Annex A/B interaction factors are not derived. (6.3.3 simplified.)
A4. Imperfection factors per curve are the textbook values and may be overridden; partial factors default to 1.0 (UK National Annex) and may be overridden. (User-editable with provenance.)
A5. section_class (Cross-section class) is taken as 1. (UB 305×165×40 S275 is class 1 in bending (Blue Book).)
A6. L_cr_y (Buckling length about y-y) is taken as 6 m. (Span 6 m.)
A7. L_cr_z (Buckling length about z-z) is taken as 1.5 m. (Compression flange restrained at 1.5 m by the deck.)
A8. k_yy (Interaction factor k_yy (member check)) is taken as 1. (No axial force; interaction factors irrelevant.)
A9. k_zz (Interaction factor k_zz (member check)) is taken as 1. (No axial force; interaction factors irrelevant.)

4. Working

[27] r_h = r
= 8.9 mm
= 8.9 mm
Root radius used by the helper

[28] A_v_rolled = A - 2 * b * t_f + (t_w + 2 * r_h) * t_f — [1](6.2.6(3))
= 51.3 cm² − 2 × 165 mm × 10.2 mm + (6 mm + 2 × 8.9 mm) × 10.2 mm
= 20.068 cm²
Rolled I shear area, load parallel to the web

[29] A_v_min = eta * (h - 2 * t_f) * t_w — [1](6.2.6(3))
= 1 × (303.4 mm − 2 × 10.2 mm) × 6 mm
= 16.98 cm²
Lower bound η h_w t_w

[30] A_v = max(A_v_rolled, A_v_min)
= max(20.068 cm², 16.98 cm²)
= 20.068 cm²
Shear area

[31] N_pl_Rd = A * f_y / gamma_M0 — [1](6.2.3(2), 6.2.4(2))
= 51.3 cm² × 275 MPa / 1
= 1410.8 kN
Plastic axial resistance

[32] W_y = W_pl_y — [1](6.2.5(2))
= 623 cm³
= 623 cm³
Section modulus about y (plastic, class 1–2)

[33] W_z = W_pl_z [1](6.2.5(2))
= 142 cm³
= 142 cm³
Section modulus about z (plastic)
[34] V_pl_Rd = A_v * f_y / (sqrt(3) * gamma_M0) [1](6.2.6(2))
= 20.068 cm² × 275 MPa / (sqrt(3) × 1)
= 318.62 kN
Plastic shear resistance
[35] u_V = V_Ed / V_pl_Rd
= 100 kN / 318.62 kN
= 0.31386
Shear utilisation
[36] rho = 0 [1](6.2.8(2))
= 0
Shear reduction factor (V_Ed ≤ 0.5 V_pl,Rd)
[37] M_y_c_Rd = (1 - rho) * W_y * f_y / gamma_M0 [1](6.2.5(2), 6.2.8)
= (1 - 0) × 623 cm³ × 275 MPa / 1
= 171.32 kN·m
Bending resistance about y
[38] M_z_c_Rd = (1 - rho) * W_z * f_y / gamma_M0 [1](6.2.5(2))
= (1 - 0) × 142 cm³ × 275 MPa / 1
= 39.05 kN·m
Bending resistance about z
[39] u_M_y = M_y_Ed / M_y_c_Rd
= 120 kN·m / 171.32 kN·m
= 0.70042
Bending utilisation, y
[40] u_M_z = M_z_Ed / M_z_c_Rd
= 0 kN·m / 39.05 kN·m
= 0
Bending utilisation, z
[41] u_N = abs(N_Ed) / N_pl_Rd
= abs(0 kN) / 1410.8 kN
= 0
Tension utilisation
[42] u_section = u_N + u_M_y + u_M_z [1](6.2.1(7))
= 0 + 0.70042 + 0
= 0.70042
Linear cross-section interaction
[45] alpha_y = imperfection factor, curve a [1](6.3.1.2)
= Designers' Guide §6.3.1
= 0.21
Imperfection factor about y
[46] alpha_z = imperfection factor, curve b [1](6.3.1.2)
= Designers' Guide §6.3.1
= 0.34
Imperfection factor about z
[47] N_cr_y = pi^2 * E * I_y / L_cr_y^2 [1](6.3.1.2(1))
= π^2 × 210 GPa × 8503 cm⁴ / (6 m)^2
= 4895.4 kN
Elastic critical force about y
[48] lambda_y = sqrt(A * f_y / N_cr_y) [1](6.3.1.2(1))
= sqrt(51.3 cm² × 275 MPa / 4895.4 kN)
= 0.53682
Non-dimensional slenderness about y

[49] Phi_y = 0.5 * (1 + alpha_y * (lambda_y - 0.2) + lambda_y^2) [1](6.3.1.2(1))
= 0.5 × (1 + 0.21 × (0.53682 - 0.2) + 0.53682^2)
= 0.67946
Φ about y
[50] chi_y = min(1, 1 / (Phi_y + sqrt(Phi_y^2 - lambda_y^2))) [1](6.3.1.2(1))
= min(1, 1 / (0.67946 + sqrt(0.67946^2 - 0.53682^2)))
= 0.91244
Reduction factor χ about y
[51] N_b_y_Rd = chi_y * A * f_y / gamma_M1 [1](6.3.1.1(3))
= 0.91244 × 51.3 cm² × 275 MPa / 1
= 1287.2 kN
Buckling resistance about y
[52] N_cr_z = pi^2 * E * I_z / L_cr_z^2 [1](6.3.1.2(1))
= π^2 × 210 GPa × 764 cm⁴ / (1.5 m)^2
= 7037.7 kN
Elastic critical force about z
[53] lambda_z = sqrt(A * f_y / N_cr_z) [1](6.3.1.2(1))
= sqrt(51.3 cm² × 275 MPa / 7037.7 kN)
= 0.44772
Non-dimensional slenderness about z
[54] Phi_z = 0.5 * (1 + alpha_z * (lambda_z - 0.2) + lambda_z^2) [1](6.3.1.2(1))
= 0.5 × (1 + 0.34 × (0.44772 - 0.2) + 0.44772^2)
= 0.64234
Φ about z
[55] chi_z = min(1, 1 / (Phi_z + sqrt(Phi_z^2 - lambda_z^2))) [1](6.3.1.2(1))
= min(1, 1 / (0.64234 + sqrt(0.64234^2 - 0.44772^2)))
= 0.90667
Reduction factor χ about z
[56] N_b_z_Rd = chi_z * A * f_y / gamma_M1 [1](6.3.1.1(3))
= 0.90667 × 51.3 cm² × 275 MPa / 1
= 1279.1 kN
Buckling resistance about z
[57] N_b_Rd = min(N_b_y_Rd, N_b_z_Rd)
= min(1287.2 kN, 1279.1 kN)
= 1279.1 kN
Governing buckling resistance
[58] u_buckling = 0 [1](6.3.1.1(1))
= 0
Flexural buckling utilisation (tension: not applicable)
[59] u_member = u_M_y + u_M_z [1](6.3.3(4))
= 0.70042 + 0
= 0.70042
Simplified member interaction

5. Results

SYMBOL	VALUE	DESCRIPTION
A_v	20.068 cm²	Shear area
N_pl_Rd	1410.8 kN	N_pl,Rd
V_pl_Rd	318.62 kN	V_pl,Rd
rho	0	ρ
M_y_c_Rd	171.32 kN·m	M_y,c,Rd

SYMBOL	VALUE	DESCRIPTION
M_z_c_Rd	39.05 kN·m	M_z,c,Rd
N_cr_y	4895.4 kN	N_cr,y
N_cr_z	7037.7 kN	N_cr,z
lambda_y	0.53682	λ̄_y
lambda_z	0.44772	λ̄_z
chi_y	0.91244	χ_y
chi_z	0.90667	χ_z
N_b_y_Rd	1287.2 kN	N_b,y,Rd
N_b_z_Rd	1279.1 kN	N_b,z,Rd
N_b_Rd	1279.1 kN	N_b,Rd
u_N	0	Axial utilisation
u_M_y	0.70042	Bending y utilisation
u_M_z	0	Bending z utilisation
u_V	0.31386	Shear utilisation
u_section	0.70042	Cross-section interaction
u_buckling	0	Buckling utilisation
u_member	0.70042	Member interaction

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Bending y M_y,Ed ≤ M_y,c,Rd	u_M_y = 0.70042	u_M_y ≤ 1	0.7	[1](6.2.5)
PASS	Cross-section interaction	u_section = 0.70042	u_section ≤ 1	0.7	[1](6.2.1(7))
PASS	Shear V_Ed ≤ V_pl,Rd	u_V = 0.31386	u_V ≤ 1	0.314	[1](6.2.6)
PASS	Tension N_Ed ≤ N_pl,Rd	u_N = 0	u_N ≤ 1	0	[1](6.2.3)
PASS	Bending z M_z,Ed ≤ M_z,c,Rd	u_M_z = 0	u_M_z ≤ 1	0	[1](6.2.5)

7. Applicability

WARNING — no-ltb: M_y_Ed: lateral-torsional buckling is not checked by this sheet; restrain the compression flange or check 6.3.2 separately.

8. Verification

Template struct.en1993-member@1.0.1 Content hash 2b5943a3c6dd7a64 Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. British Standards Institution. Eurocode 3: Design of Steel Structures—Part 1-1: General Rules and Rules for Buildings. BS EN 1993-1-1. BSI.
2. Gardner L, Nethercot DA. Designers' Guide to EN 1993-1-1: Eurocode 3: Design of Steel Structures, General Rules and Rules for Buildings. Thomas Telford; 2005.